

Eric Richter (EN/FR/DE/SV)

Aspiring Leader in the European Energy Transition

Residence: Coppet, Switzerland | Citizenship: Swedish (EU/EFTA), Swiss

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EXPERIENCE

GE Vernova | Birr, Switzerland

Product Quality Engineer Intern (operations)

September 2025–January 2026

- Managed continuous improvement initiatives in the lean “Built-in Quality” transformation across 4 turbine parts production lines.
- Facilitated a project mandate for external logistics solution providers & led a cross-functional root cause analysis (RCA) team.
- Optimized a transport process cycle time by 80% & participated in 3 Kaizen events.

Boston University | Boston, MA USA

Research Fellow at the Lubner Group

August 2024–June 2025

- Modeled mass & heat transport for carbon capture and thermal wave sensing using a custom Python framework and COMSOL.
- Created a framework to synthesize modeled kinetic behaviors into key research insights.

Sensirion AG | Stäfa, Switzerland

R&D Intern in the Temperature and Humidity Sensors Team

September 2023–February 2024

- Engineered, built, characterized and iterated on a high-precision humidity sensor, costing 1% of the alternatives.
- Reduced stabilization time by 50% with respect to reference R&D equipment using integrated control in C, C++, Arduino.

ETH | Zürich, Switzerland

Research Assistant at EIS in the Institute for Neuro-Informatics

February 2023–June 2023

- Assessed an evolutionary method for training neuromorphic networks that perform AI tasks on the edge of hardware.
- Incorporated small-world graph theory considerations to promote formation of clusters of highly connected neurons.

EPFL | Lausanne, Switzerland

Research Assistant at LMIS1

February 2022–June 2022

- Optimized a class of thermally sensitive electro-magnetically coupled superconducting resonators using COMSOL EM library.
- Achieved quality factors of 5000, and generated 15+ physical design insights.

EPFL | Lausanne, Switzerland

Research Assistant at MOBOTS - Hiveopolis

September 2021–January 2022

- Delivered a fully Python automated robotic laser doppler vibration analysis platform dividing characterization time by 60.

EPFL | Lausanne, Switzerland

Teaching Assistant in Mathematical Analysis

February 2021–July 2021

EDUCATION

ETH | Zürich, Switzerland

Master of Science (MSc) in Mechanical Engineering (GPA 5.5/6)

August 2025

EPFL | Lausanne, Switzerland

Master of Science (MSc) 61 ECTS in Microengineering (GPA 5.3/6)

July 2022

EPFL | Lausanne, Switzerland

Bachelor of Science (BSc) in Microengineering (GPA 5.23/6)

July 2021

USJ | Beirut, Lebanon

Off-Curriculum International Relations Studies

December 2017

LEADERSHIP

Energy Student Association (EMC²) | Zürich, Switzerland

Panel Discussion Host, Event Organizer

January 2024–May 2024

- Coordinated and moderated a policy debate about carbon pricing in front of a 200+ person audience.

ShARE EPFL | Lausanne, Switzerland

Interim Vice-President

June 2022–November 2022

Mentor, Consultant

September 2020–June 2022

- Coordinated the association & completed “ShARE’s Leadership Programme of Excellence”, a social impact focused program.
- Collaborated with EPFL’s startup accelerator by completing the product-market fit analysis for 3 startups.

iGEM Synthetic Biology Competition | Cambridge, MA (USA)

Head Engineer, Gold Medal, Best Website, Best Environmental Project Nomination

February 2021–November 2021

- Prototyped of a reactor for bioremediation of heavy metals and participated in bio- and chemical- synthetic biology experiments.
- Involved 3 stakeholders to guide our value proposition & presented the problem in 2 high-schools and in a 4-episode podcast.

SKILLS

Languages: English (Native), French (Native), German (Proficient), Swedish (Conversational).

Programming: Python, C, C++, MATLAB/Simulink, Java, Javascript, VHDL, ARM Assembly.

Engineering Software: CAD (CATIAV5, fusion360, Onshape), Multiphysics (COMSOL), Circuit (LTSpice), FPGA (Vivado Suite).

Engineering Focus: Control Theory, Semi-Conductor Technologies, Sensor Technologies, Electronics.

Operations: SAP, SQDC, Quality Management (ISO9001-2015 QMS, FMEA, 8D, RCA), Lean (5S, Kaizen, VSM, JIT, Flow).